

Biexciton Binding and Radiative Lifetimes in Strongly Oblate Semi-Ellipsoidal Quantum Dots

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Abstract

We investigate the ground exciton and the biexciton state in a hard-wall, strongly oblate semi-ellipsoidal GaAs quantum dot. Correlated variational wave functions are evaluated using a ground-orbital-matched importance transformation and adaptive quasi-Monte Carlo integration. To avoid numerical bias from independently evaluated symmetry-reduced pieces, the final biexciton estimator retains the complete symmetrized correlation factor, all six Coulomb terms, and all four kinetic contributions, removing only the redundant global azimuthal rotation. A safeguarded two-budget coordinate pattern search is used to optimize the variational parameters at seven representative geometries. Increasing the axial semiaxis from $0.5r_B$ to $2r_B$ at fixed lateral semiaxis $5r_B$ lowers the corrected biexciton energy from 89.65 Ry to 3.67 Ry, whereas increasing the lateral semiaxis from $3r_B$ to $10r_B$ at fixed axial semiaxis r_B changes it only from 21.82 Ry to 20.09 Ry. The central biexciton binding energy is positive throughout the sampled range and decreases monotonically as either dimension increases. Binding is resolved at all geometries except $(a, c) = (10, 1)r_B$, where $E_{\text{bind}} = 0.031(20)$ Ry is only marginally positive at the present numerical resolution. Enlargement of the dot increases the exciton coincidence overlap and oscillator strength, reducing the calculated exciton lifetime from 2.548 ps to 1.437 ps along the axial scan and from 2.767 ps to 1.068 ps along the lateral scan. Under the adopted radiative-channel convention, $\tau_{XX} = \tau_X/4$. These results separate the dominant axial control of the absolute confinement energy from the pronounced lateral control of weak biexciton binding and radiative strength.

Keywords: semi-ellipsoidal quantum dot; exciton; biexciton; binding energy; variational method; quasi-Monte Carlo integration; radiative lifetime

1 Introduction

Semiconductor quantum dots provide discrete, atom-like carrier spectra while retaining substantial control through composition, confinement, and external geometry. Their neutral exciton (X) and biexciton (XX) states form the elementary ladder for correlated photon emission. In particular, the biexciton–exciton cascade has enabled regulated photon-pair emission [1, 2] and polarization-entangled photon generation [3]. The biexciton binding energy determines the separation of the two cascade transitions and therefore affects spectral addressability, resonant excitation, and regimes in which the emitted photons approach equal energies.

The energy of a confined biexciton is not fixed by a single Coulomb scale. Electron–hole attraction competes with electron–electron and hole–hole repulsion, while the carrier masses and the confinement geometry determine how strongly the four particles can rearrange. Consequently, biexcitons may be bound, antibound, or close to zero binding even within the same material system. Experimental studies have likewise shown that biexciton binding can change strongly with confinement and dot structure [4, 5]. This sensitivity makes the biexciton a useful probe of few-particle correlations, but it also makes its numerical evaluation demanding: the binding energy is a small difference between two much larger correlated energies.

Strongly oblate quantum dots are especially suitable for separating the roles of axial and lateral confinement. Their two semiaxes provide independent geometrical controls, while the large separation between axial and in-plane energy scales permits an adiabatic treatment of the single-particle states. Variational calculations for strongly oblate ellipsoidal dots have demonstrated geometry-dependent energies, binding energies, recombination energies, and radiative lifetimes [6], and subsequent work extended the same framework to nonlinear optical response [7]. The semi-ellipsoidal boundary considered here is relevant to flattened or dome-like dots and has previously been used for single-particle and interband absorption calculations [8]. Unlike a full ellipsoid, the dot occupies only one side of the basal plane, which changes the axial boundary condition while retaining axial rotational symmetry.

The purpose of this work is to determine how the lateral semiaxis a and axial semiaxis c control the corrected energy, binding, and radiative lifetime of the biexciton in a strongly oblate semi-ellipsoidal GaAs dot. We restrict the study to the single-particle ground configuration and use correlated variational states for both X and XX . The high-dimensional integrals are evaluated after a state-matched importance transformation. For the final biexciton calculation we retain the full symmetrized correlation expression and remove only the redundant global rotation, thereby avoiding the accumulation of independent quadrature errors among analytically equivalent reduced terms. The variational parameters are then refined by a safeguarded coordinate pattern search that verifies accepted moves at two integration budgets.

Seven geometries form two intersecting scans: $c = 0.5, 1, 1.5, 2$ at fixed $a = 5$, and $a = 3, 5, 7, 10$ at fixed $c = 1$, with all lengths measured in the effective electron Bohr radius. From the optimized states we obtain E_X , E_{XX} , $E_{\text{bind}} = 2E_X - E_{XX}$, the exciton coincidence overlap, and the radiative lifetimes τ_X and τ_{XX} . Particular attention is paid to the numerical resolution of the small binding energy. The reported error bars are rough propagations of the adaptive integrator's internal estimates rather than formal statistical confidence intervals.

2 Model and theoretical framework

The interacting exciton and biexciton ground-state energies are obtained with the variational method: trial correlation factors are optimized by minimizing the corresponding Hamiltonian expectation values. We first establish the single-particle confinement states on which these trial states are built.

2.1 Geometry, units, and single-particle ground state

We consider a hard-wall semi-ellipsoidal quantum dot with cylindrical coordinates $\mathbf{r} = (r, \phi, z)$ and confinement potential

$$V_{\text{conf}}(r, z) = \begin{cases} 0, & r^2/a^2 + z^2/c^2 \leq 1 \text{ and } z \geq 0, \\ \infty, & \text{otherwise,} \end{cases} \quad c/a \ll 1. \quad (1)$$

The in-plane and axial semiaxes are a and c , respectively, as illustrated in Fig. 1.

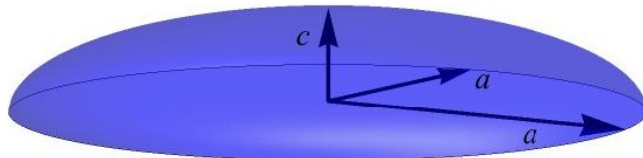


Figure 1: Semi-ellipsoidal quantum-dot geometry, with lateral semiaxis a and axial semiaxis c .

Lengths and energies are expressed in the effective electron Bohr radius and Rydberg,

$$r_B = \frac{4\pi\epsilon_0\epsilon_r^{(E)}\hbar^2}{m_e e^2}, \quad E_R = \frac{\hbar^2}{2m_e r_B^2}. \quad (2)$$

For the GaAs energy calculation we use the standard bulk parameter set [9], $\epsilon_r^{(E)} = 12.8$, $m_e = 0.067m_0$, and $m_h = 0.45m_0$, giving $r_B = 10.109\,654\,\text{nm}$ and $E_R = 5.563\,851\,565\,\text{meV}$.

Because the axial confinement is much stronger than the in-plane confinement, the single-particle problem is treated adiabatically [8]. At fixed r , the available height is

$$z_{\max}(r) = c\sqrt{1 - r^2/a^2}, \quad (3)$$

and the lowest axial factor is

$$\chi_1(z; r) = \sqrt{\frac{2}{z_{\max}(r)}} \sin \left[\frac{\pi z}{z_{\max}(r)} \right]. \quad (4)$$

For a carrier of mass m_q with $q = e, h$ the axial eigenenergy acts as the adiabatic potential for the slow planar motion,

$$U_q(r) = \frac{m_e}{m_q} \frac{\pi^2}{z_{\max}^2(r)} = \frac{m_e}{m_q} \frac{\pi^2}{c^2} \left(1 + \frac{r^2}{a^2} + O(r^4/a^4) \right), \quad (5)$$

where lengths and energies are in the effective units of Eq. (2). Thus, after subtracting the constant axial energy, the planar harmonic-oscillator Hamiltonian is

$$H_{q,\parallel} = \frac{m_e}{m_q} \left[-\nabla_{\parallel}^2 + \frac{\pi^2 r^2}{a^2 c^2} \right]. \quad (6)$$

Writing $\lambda = \pi/(ac)$, its normalized ground-state component and the corresponding axial and planar ground-state energies are

$$\varphi_{q,\parallel}(r) = \sqrt{\frac{\lambda}{\pi}} e^{-\lambda r^2/2}, \quad \varepsilon_{q,z} = \frac{m_e}{m_q} \frac{\pi^2}{c^2}, \quad \varepsilon_{q,\parallel} = \frac{m_e}{m_q} \frac{2\pi}{ac}. \quad (7)$$

The total ground-state confinement energy of carrier q is therefore

$$E_q = \varepsilon_{q,z} + \varepsilon_{q,\parallel} = \frac{m_e}{m_q} \left(\frac{\pi^2}{c^2} + \frac{2\pi}{ac} \right). \quad (8)$$

The normalized carrier ground orbital is therefore $\psi_q(r, \phi, z) = \chi_1(z; r)\varphi_{q,\parallel}(r)$. Thus the uncorrelated exciton and biexciton confinement energies are

$$E_X^{(0)} = E_e + E_h, \quad E_{XX}^{(0)} = 2E_e + 2E_h = 2E_X^{(0)}. \quad (9)$$

The band gap is not included in these energies.

2.2 Correlated exciton

We denote electron coordinates by $1, 2$ and hole coordinates by a, b ; the ground exciton uses coordinates 1 and a . Pair distances are the three-dimensional Euclidean distances $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. In the effective units of Eq. (2), its Hamiltonian is

$$H_X = h_e(1) + h_h(a) - \frac{2}{r_{1a}}, \quad h_e\psi_e = E_e\psi_e, \quad h_h\psi_h = E_h\psi_h. \quad (10)$$

The uncorrelated state $\Psi_{0X} = \psi_e(1)\psi_h(a)$ is therefore an eigenstate of $h_e + h_h$ with energy $E_X^{(0)} = E_e + E_h$. Electron-hole correlation is introduced through

$$\Psi_X = \Psi_{0X} F_X, \quad F_X = \exp(-\alpha_X r_{1a}), \quad (11)$$

where $\alpha_X > 0$ is varied independently at each geometry. The variational energy is evaluated from

$$E_X = \frac{\langle \Psi_X | H_X | \Psi_X \rangle}{\langle \Psi_X | \Psi_X \rangle}. \quad (12)$$

The denominator is

$$N_X = \int |\Psi_{0X}|^2 F_X^2 d\tau = \int |\Psi_{0X}|^2 e^{-2\alpha_X r_{1a}} d\tau. \quad (13)$$

Using the single-particle eigenvalue equations in Eq. (10) and integrating the kinetic terms by parts gives

$$E_X = E_X^{(0)} + \frac{1}{N_X} \int |\Psi_{0X}|^2 \left[|\nabla_1 F_X|^2 + \frac{m_e}{m_h} |\nabla_a F_X|^2 - \frac{2F_X^2}{r_{1a}} \right] d\tau. \quad (14)$$

Since $|\nabla_1 r_{1a}| = |\nabla_a r_{1a}| = 1$, the kinetic contribution generated by F_X is

$$K_X(\alpha_X) = \alpha_X^2 \left(1 + \frac{m_e}{m_h} \right). \quad (15)$$

Defining the Coulomb numerator as

$$V_X = \int |\Psi_{0X}|^2 F_X^2 \left(-\frac{2}{r_{1a}} \right) d\tau \quad (16)$$

reduces Eq. (14) to

$$E_X = E_X^{(0)} + \Delta E_X, \quad \Delta E_X = K_X + \frac{V_X}{N_X}. \quad (17)$$

For each dot geometry, the remaining integrals N_X and V_X are evaluated numerically.

2.3 Correlated biexciton

The two-electron–two-hole interaction in effective units is

$$V_{XX} = 2 \left(\frac{1}{r_{12}} + \frac{1}{r_{ab}} - \frac{1}{r_{1a}} - \frac{1}{r_{1b}} - \frac{1}{r_{2a}} - \frac{1}{r_{2b}} \right). \quad (18)$$

For the biexciton we use the ansatz []

$$\Psi_{XX} = \Psi_{0XX} F_{XX}, \quad \Psi_{0XX} = \psi_e(1)\psi_e(2)\psi_h(a)\psi_h(b), \quad (19)$$

with the exchange-symmetric correlation factor

$$\begin{aligned} F_{XX} &= G(P + Q), \\ G &= r_{ab}^\gamma e^{-\delta r_{ab}}, \\ P &= \exp[-\alpha(r_{1a} + r_{2b}) - \beta(r_{1b} + r_{2a})], \\ Q &= \exp[-\beta(r_{1a} + r_{2b}) - \alpha(r_{1b} + r_{2a})]. \end{aligned} \quad (20)$$

The four variational parameters $(\alpha, \beta, \gamma, \delta)$ control the direct and cross electron–hole pairings and the hole–hole correlation.

To write the kinetic numerator compactly, define $\mathbf{g}_i^P = \nabla_i \ln(GP)$ and $\mathbf{g}_i^Q = \nabla_i \ln(GQ)$. For electron coordinates the derivatives of G vanish, whereas for hole coordinates they contain the logarithmic derivative of G in Eq. (20). Unlike the exciton kinetic contribution, the biexciton kinetic numerator has no analytical form and is written as

$$K_{XX} = \int |\Psi_{0XX}|^2 G^2 \left[\sum_{i=1,2} |P\mathbf{g}_i^P + Q\mathbf{g}_i^Q|^2 + \frac{m_e}{m_h} \sum_{j=a,b} |P\mathbf{g}_j^P + Q\mathbf{g}_j^Q|^2 \right] d\tau. \quad (21)$$

The energy is then

$$E_{XX} = E_{XX}^{(0)} + \Delta E_{XX}, \quad \Delta E_{XX} = \frac{K_{XX} + \langle \Psi_{XX} | V_{XX} | \Psi_{XX} \rangle}{\langle \Psi_{XX} | \Psi_{XX} \rangle}. \quad (22)$$

In the numerical calculation, the kinetic contribution in Eq. (21) and the potential contribution are combined and evaluated in a single integral. No direct/cross or particle-exchange multiplicities are introduced in the final estimator.

We define the biexciton binding energy as

$$E_{\text{bind}} = 2E_X - E_{XX} = 2\Delta E_X - \Delta E_{XX}. \quad (23)$$

With this convention $E_{\text{bind}} > 0$ denotes a bound biexciton. The single-particle part cancels exactly, making Eq. (23) a small difference of correlation corrections.

2.4 Radiative lifetimes

Following Hayrapetyan *et al.* [6], we calculate the exciton and biexciton radiative lifetimes from

$$\tau_X = \frac{2\pi\epsilon_0 m_e c_0^3 \hbar^2}{\sqrt{\epsilon_r^{(\text{opt})}} e^2 E_{\text{exc}}^2 f_X}, \quad \tau_{XX} = \frac{\tau_X}{4}. \quad (24)$$

The relation $\tau_{XX} = \tau_X/4$ follows the biexciton decay-channel convention of Ref. [6]; it is not an independently optimized quantity.

Evaluation of τ_X requires the oscillator strength and transition energy, which are

$$f_X = \frac{E_P}{E_{\text{exc}}} |M_X|^2, \quad E_{\text{exc}} = E_g + E_X, \quad (25)$$

where, E_g is the band gap. The remaining quantity M_X is the electron-hole coincidence overlap. At coincidence the exciton Jastrow factor in Eq. (11) is unity, and because the electron and hole ground orbitals are matched in the present model, the normalized overlap reduces to

$$M_X = \frac{1}{\sqrt{N_X}}, \quad |M_X|^2 = \frac{1}{N_X}. \quad (26)$$

Thus the lifetime can be obtained from the already calculated exciton norm; no additional oscillator-strength integral is required.

For the optical calculation we take $E_g = 1.5192$ eV, $m_e = 0.0665m_0$, $E_P = 22.71$ eV, and $\epsilon_r^{(\text{opt})} = 12.91$, matching the parameter set of Ref. [6]. We keep this optical set distinct from the bulk parameters in Eq. (2) because the latter define the units and single-particle states used in the variational energy calculation, whereas the former enter the dimensional radiative-lifetime parametrization. This separation preserves the energy calculation while reproducing the cited optical calibration.

3 Numerical method and validation

3.1 Geometry and integration parametrization

We study the seven geometries listed in Table 1. The common point (5, 1) links a scan of axial confinement to a scan of lateral confinement.

Direct integration in the physical coordinates is inefficient because the single-particle orbitals occupy only a small part of the nominal domain. We therefore transform every particle to variables matched to the ground-state density. The axial coordinate is first written

$$z = u c \sqrt{1 - r^2/a^2}, \quad 0 < u < 1. \quad (27)$$

Table 1: Quantum-dot semiaxes used in the axial and lateral scans, in effective Bohr radii.

Scan	a	c
Axial	5	0.5, 1, 1.5, 2
Lateral	3, 5, 7, 10	1

The numerical radial domain is restricted to the validated cutoff $R = 0.9a$. Using the ground-state in-plane length ℓ , the reference radial and axial densities are

$$p_r(r) = \frac{1}{C} \frac{2r}{\ell^2} e^{-r^2/\ell^2}, \quad C = 1 - e^{-R^2/\ell^2}, \quad p_u(u) = 2 \sin^2(\pi u). \quad (28)$$

Both reference densities have analytical cumulative distributions. For the radial density, the inverse is

$$r(v) = \ell \sqrt{-\ln(1 - Cv)}, \quad 0 < v < 1, \quad (29)$$

whereas the axial cumulative distribution has no elementary closed form inverse and is tabulated once and interpolated. This transformation cancels most of the localized single-particle density and leaves an order-unity transformed integrand. It is an importance transformation, not a symmetry reduction.

The final estimator removes only the redundant global azimuthal rotation. For the exciton, the hole angle is fixed to zero, the electron angle spans $[0, 2\pi]$, and a factor 2π restores the global rotation. For the biexciton, $\phi_a = 0$ while ϕ_1 , ϕ_2 , and ϕ_b are each integrated independently over $[0, 2\pi]$, again with an overall factor 2π . No reflection folding or particle-exchange multiplicities are used. The complete $G(P + Q)$ factor, the six Coulomb interactions in Eq. (18), and the four kinetic contributions in Eq. (21) remain in the pointwise integrand.

3.2 Adaptive quasi-Monte Carlo quadrature

Every integral is evaluated with the Wolfram Language implementation of adaptive quasi-Monte Carlo quadrature. Low-discrepancy points improve the coverage of the high-dimensional unit hypercube relative to independent pseudorandom sampling [10], while adaptive subdivision directs additional evaluations toward regions with larger estimated error. The production settings are

$$\begin{aligned} \text{Method} &= \text{AdaptiveQuasiMonteCarlo}, \\ \text{BisectionDithering} &= 0, \\ \text{AccuracyGoal} &= \infty, \quad \text{PrecisionGoal} = 4. \end{aligned} \quad (30)$$

The stage-dependent point budget is supplied through `MaxPoints`. A finite absolute accuracy goal is inappropriate because some biexciton numerator and norm integrals are much smaller than a typical absolute tolerance. Reaching `MaxPoints` therefore indicates that the requested relative goal was not attained before the budget cap; the returned integral value and internal error estimate are nevertheless retained.

The Hamiltonian numerator and norm denominator are independent adaptive calls. They do not share a fixed Sobol point set. This preserves separate adaptation to their different integrands. The internal quadrature errors returned for the numerator and denominator are treated as independent and propagated to the exciton and biexciton energies, the binding energy, and the radiative lifetimes. The resulting error bars are numerical-resolution indicators, not statistical confidence intervals: they neglect covariance between the adaptive integrals, possible systematic quasi-Monte Carlo bias, and uncertainty in the location of the variational optimum.

The final estimator was selected through a sequence of checks at (5, 1). These included the exact exciton normalization at $\alpha_X = 0$, integration budget comparisons, dithered adaptive partitions, a complete four-angle biexciton calculation, and the final global-rotation-fixed calculation.

The two angular formulations agree within the observed resolution, while removing the redundant global angle reduces the numerical dimension. Earlier direct/cross and particle-relabeling identities remain analytically valid, but their independent numerical evaluations exhibited percent-level differences and were therefore not used to assemble the final energies.

3.3 Variational search

The variational search proceeded in two phases. The first was an exploratory multidimensional search performed with the Wolfram Language function `NMinimize` using the Nelder–Mead method. It varied the parameters simultaneously at a lower integration budget and located approximate minima for each geometry. Because small objective differences can be dominated by quasi-Monte Carlo fluctuations, these first-phase optima were used only as starting points.

The second phase refined those starting points with an automated local coordinate pattern search using the integration setup of [section 3.1](#). Positive and negative displacements of each parameter were screened at a moderate budget; the center and the most promising candidates were then recomputed at a larger budget, with progressively smaller displacements used around stable candidates. Automation generated and evaluated the candidates, but final selection also relied on inspecting the energies, integration-error estimates, stability between budgets, neighboring points in parameter space, and consistency of the geometry trends. A parameter set was retained only when this review indicated a stable local minimum rather than an isolated low numerical fluctuation.

4 Results and discussion

4.1 Corrected exciton and biexciton energies

The correlation corrections and corrected energies are collected in [Table 2](#) of the Appendix. All energies are reported in the effective Rydberg of [Eq. \(2\)](#); conversion to dimensional units is used only for the radiative-lifetime calculation. The values include the single-particle confinement and correlation corrections but exclude the band gap. Because the zeroth-order energies are analytical, each correlation correction has the same numerical integration error as its corresponding corrected energy. Although these rough errors are retained in the table, they are too small to be visible on the scale of the biexciton-energy curves in [Fig. 2](#).

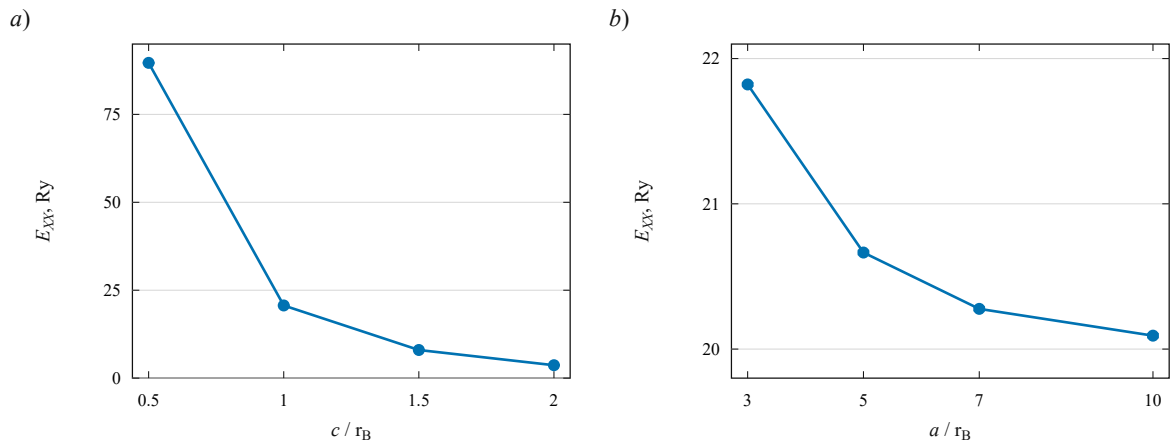


Figure 2: Corrected biexciton energy as a function of (a) the axial semiaxis c at fixed $a = 5r_B$ and (b) the lateral semiaxis a at fixed $c = r_B$, in effective Rydbergs. The band gap is excluded. Lines are guides to the eye, and numerical error bars are smaller than the symbols at the displayed scales.

Along the axial scan, E_{XX} decreases steeply and nonlinearly, with most of the change occurring between the two most strongly confined geometries. This behavior reflects the strong sensitivity of the confinement energy to the short semiaxis in the oblate regime.

The lateral scan is much flatter: enlarging the in-plane semiaxis produces only a modest reduction in E_{XX} . The axial dimension is therefore the dominant geometrical control of the absolute biexciton energy over the sampled range, consistent with the separation of axial and lateral energy scales in Eq. (7).

4.2 Biexciton binding

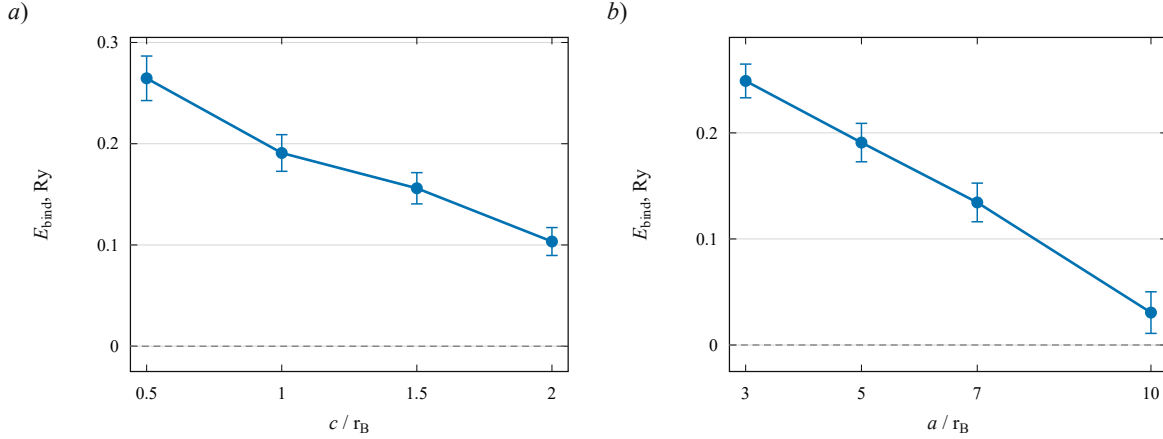


Figure 3: Binding energy $E_{\text{bind}} = 2E_X - E_{XX}$ as a function of (a) c at fixed $a = 5r_B$ and (b) a at fixed $c = r_B$, in effective Rydbergs. Positive values correspond to a bound biexciton. The dashed line marks zero binding. Error bars are rough first-order propagations of the adaptive-integration estimates, not statistical confidence intervals.

The central binding energy is positive throughout the sampled geometries and decreases monotonically when either semiaxis is increased (Fig. 3). Thus relaxing axial or lateral confinement reduces the net correlation advantage of the four-particle state relative to two isolated excitons. The lateral scan approaches the zero-binding threshold more closely than the axial scan.

The binding is well separated from zero at every geometry except the widest lateral dot. There, the numerical-resolution estimate is comparable to the small central binding energy. Because that estimate omits systematic integration bias and uncertainty in the optimum, we classify the point as marginally positive rather than claiming firmly resolved binding. No sign change is demonstrated within the studied range.

Because the single-particle part cancels in Eq. (23), the origin of this trend is exposed by comparing the two correlation corrections in Fig. 4. Both become less negative as the dot is enlarged, while the separation between $2\Delta E_X$ and ΔE_{XX} narrows. That separation is precisely E_{bind} , so the convergence of the two curves accounts directly for the suppression of binding, particularly along the lateral scan.

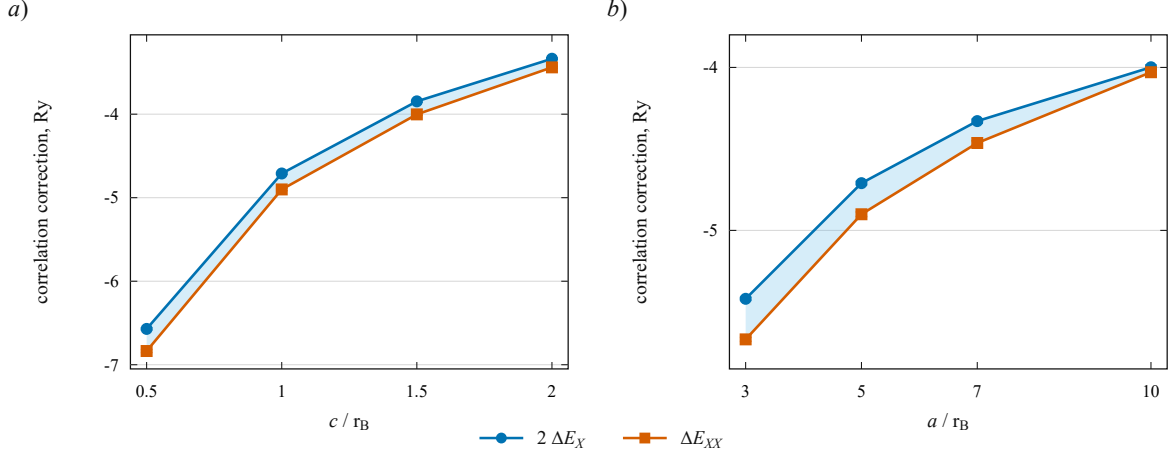


Figure 4: Comparison of the two-exciton correction $2\Delta E_X$ with the biexciton correction ΔE_{XX} as functions of (a) c at fixed $a = 5r_B$ and (b) a at fixed $c = r_B$, in effective Rydbergs. Their vertical separation equals the binding energy. Lines are guides to the eye; numerical error bars are smaller than the plotted symbols at this scale.

4.3 Variational parameters and optical intermediates

The complete optimized biexciton parameters are reported in Table 3 of the Appendix and summarized graphically in Fig. 5. The exciton parameter α_X generally decreases along the axial scan and increases along the lateral scan. For the biexciton, the optimized cross-pair exponent β becomes very small at the larger axial sizes. The composite quantity γ/δ , which sets the maximum of the isolated factor $r_{ab}^\gamma e^{-\delta r_{ab}}$, increases smoothly as either dot dimension is enlarged. This indicates that the hole-hole factor shifts toward larger characteristic separations in larger dots.

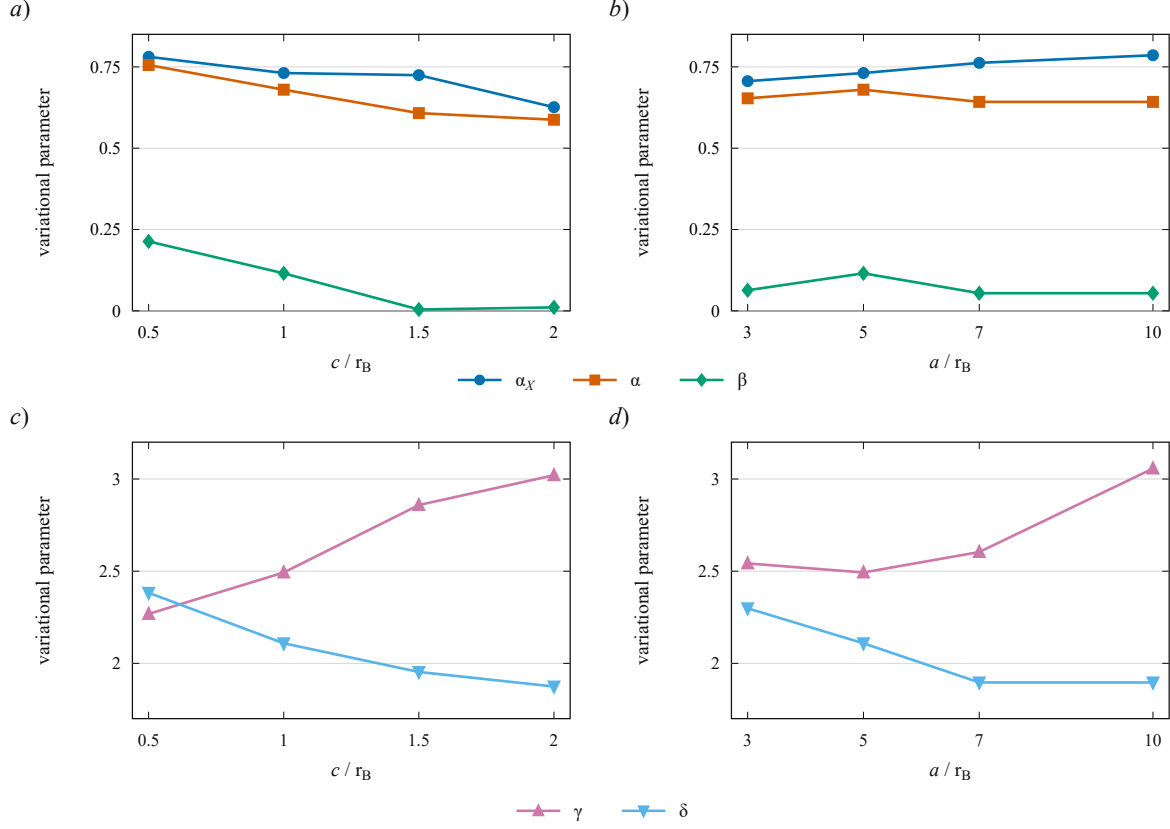


Figure 5: Optimized variational parameters along the axial scan at fixed $a = 5r_B$ and the lateral scan at fixed $c = r_B$. Panels (a) and (b) show α_X , α , and β ; panels (c) and (d) show γ and δ . Lines are guides to the eye.

These parameter-level trends should be interpreted more cautiously than the energies. The objective is numerically rough, the pattern search is local, and different parameter combinations can produce energies separated by less than the practical quadrature resolution. The parameters characterize the retained variational minimum but are not claimed to be unique.

The coincidence overlap is shown in Fig. 6, while the derived optical quantities are listed in Table 4 of the Appendix. The overlap increases as either dot dimension is enlarged. It tends toward saturation at the larger axial sizes, whereas it continues to rise across the lateral scan. The oscillator strength follows the same behavior.

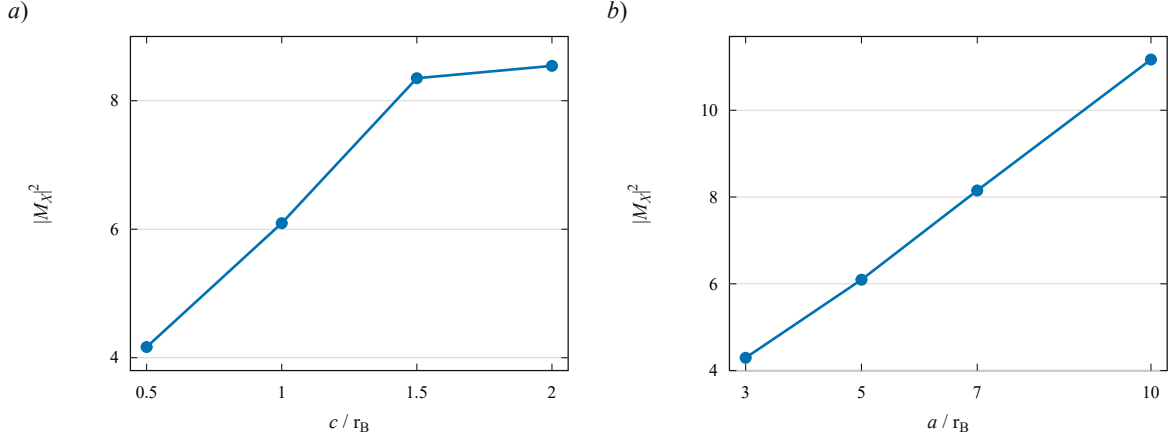


Figure 6: Squared exciton coincidence overlap $|M_X|^2$ as a function of (a) c at fixed $a = 5r_B$ and (b) a at fixed $c = r_B$. Lines are guides to the eye, and numerical error bars are smaller than the plotted symbols.

4.4 Radiative lifetimes

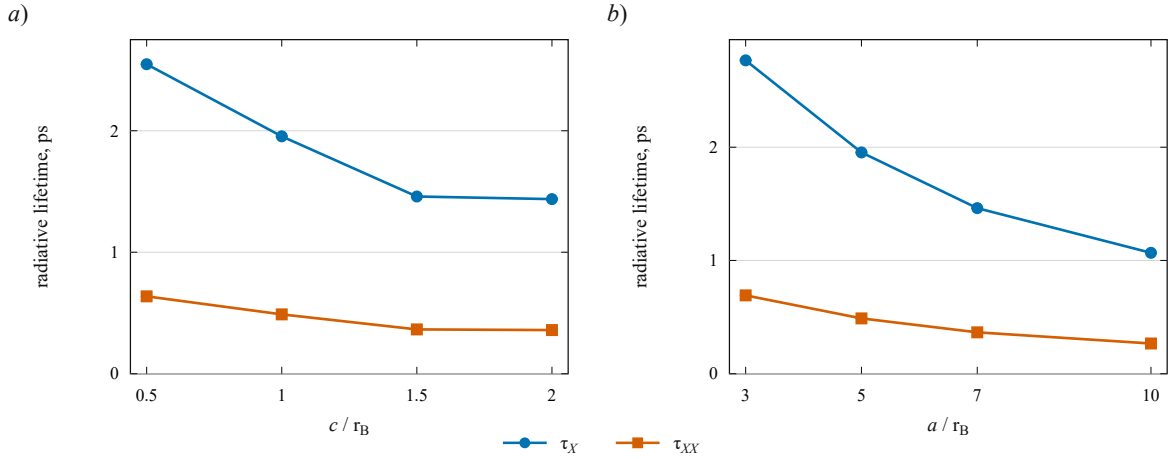


Figure 7: Exciton and biexciton radiative lifetimes as functions of (a) c at fixed $a = 5r_B$ and (b) a at fixed $c = r_B$. The values use the corrected optical transition energy and the coincidence overlap. The adopted decay-channel convention gives $\tau_{XX} = \tau_X/4$. Propagated error bars are smaller than the plotted symbols.

Both radiative lifetimes decrease as the dot is enlarged (Fig. 7). Along the axial scan, the decrease weakens and becomes nearly saturated for the largest dots. Along the lateral scan, the decline remains pronounced across the sampled range. The biexciton lifetime follows the same geometry dependence at one quarter of the exciton value.

Relaxing confinement decreases E_{exc} , which alone would tend to increase the lifetime. The simultaneous increase in $|M_X|^2$ is stronger, however. Since substitution of Eq. (25) into Eq. (24) gives $\tau_X \propto (E_{\text{exc}}|M_X|^2)^{-1}$, the overlap enhancement dominates and the lifetime decreases.

4.5 Comparison and limitations

The present geometry trends are qualitatively consistent with calculations for full strongly oblate ellipsoidal dots, where the semiaxes tune the correlated energies and binding and the exciton overlap sets the radiative scale [6, 7]. A direct numerical comparison is not appropriate, however,

because the semi-ellipsoidal boundary changes the axial single-particle state, and the present four-particle trial function and full integration protocol differ from those earlier calculations.

The model assumes impenetrable barriers, bulk effective masses, static dielectric screening for the energy calculation, and an adiabatic single-particle ground state. Finite band offsets, dielectric mismatch, strain, atomistic symmetry, and phonon-assisted decay are outside the present scope. The lifetime calculation follows the optical parameter set and decay-channel convention of Ref. [6]; it should therefore be interpreted as the radiative lifetime within that model rather than as a complete experimental decay time.

Numerically, the reported bars quantify only first-order propagation of the adaptive integrator’s internal errors. The smooth trends and the positive binding away from the widest lateral dot are robust on that scale, but a higher-resolution targeted calculation would be required to determine firmly whether the marginal endpoint is bound or antibound.

5 Conclusions

We have calculated the ground exciton and biexciton state of a strongly oblate semi-ellipsoidal GaAs quantum dot for seven representative geometries. A ground-orbital-matched importance transformation, a full global-rotation-fixed biexciton estimator, and a safeguarded two-budget pattern search were combined to limit the numerical sensitivity of the small binding energy.

The corrected biexciton energy decreases as either semiaxis increases, but the axial dimension controls the absolute energy much more strongly. At fixed $a = 5r_B$, increasing c from $0.5r_B$ to $2r_B$ reduces E_{XX} by a factor of 24.4, whereas increasing a from $3r_B$ to $10r_B$ at fixed $c = r_B$ changes it by only 7.9%.

The central binding energy remains positive throughout the sampled range and decreases monotonically as the dot is enlarged. It is resolved as positive by the present rough numerical diagnostic at every geometry except (10, 1). At that largest lateral size, $E_{\text{bind}} = 0.031(20)$ Ry is only marginally positive, and no firm sign claim is made.

Increasing either semiaxis also enhances the exciton coincidence overlap and oscillator strength. This effect outweighs the reduction of optical transition energy and shortens both radiative lifetimes. The calculated exciton lifetime ranges from 2.767 ps to 1.068 ps; the adopted four-channel relation gives $\tau_{XX} = \tau_X/4$.

The results therefore identify distinct geometrical controls: the short axial dimension sets the dominant confinement-energy scale, while lateral enlargement provides a strong route for suppressing weak biexciton binding and enhancing radiative strength. Extensions to finite barriers, strained or atomistic confinement, and a targeted high-resolution study of the near-zero-binding regime are natural next steps.

Data availability

The numerical source files, checkpointed variational searches, machine-readable result table, and plotting notebook used in this work are retained in the project repository. A public repository or archival DOI will be supplied upon publication.

Author contributions

To be completed after the author list and journal contribution taxonomy are finalized.

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A Supplementary numerical information

This appendix collects the numerical tables supporting the main-text figures and the estimator-selection record. [Table 2](#) gives the correlation corrections, corrected energies, and binding energies.

Table 2: Exciton and biexciton correlation corrections, corrected energies, and binding energy, all in effective Rydbergs. Errors are rough propagated numerical integration estimates.

a	c	ΔE_X	E_X	ΔE_{XX}	E_{XX}	E_{bind}
5	0.5	-3.286 ± 0.007	44.958 ± 0.007	-6.836 ± 0.018	89.652 ± 0.018	0.265 ± 0.022
5	1	-2.355 ± 0.006	10.428 ± 0.006	-4.901 ± 0.014	20.665 ± 0.014	0.191 ± 0.018
5	1.5	-1.923 ± 0.005	4.079 ± 0.005	-4.002 ± 0.011	8.002 ± 0.011	0.156 ± 0.015
5	2	-1.668 ± 0.005	1.889 ± 0.005	-3.440 ± 0.010	3.674 ± 0.010	0.103 ± 0.014
3	1	-2.710 ± 0.005	11.035 ± 0.005	-5.669 ± 0.012	21.822 ± 0.012	0.249 ± 0.016
7	1	-2.165 ± 0.006	10.206 ± 0.006	-4.464 ± 0.013	20.277 ± 0.013	0.134 ± 0.018
10	1	-1.999 ± 0.007	10.061 ± 0.007	-4.029 ± 0.014	20.092 ± 0.014	0.031 ± 0.020

[Table 3](#) lists the final parameters retained by the safeguarded pattern search. The ratio γ/δ is also shown because it locates the maximum of the isolated hole-hole factor $r_{ab}^\gamma e^{-\delta r_{ab}}$.

Table 3: Final optimized parameters for the exciton and biexciton states.

a	c	α_X	α	β	γ	δ	γ/δ
5	0.5	0.780939	0.755694	0.213328	2.268187	2.382724	0.952
5	1	0.731089	0.679864	0.115335	2.492963	2.108847	1.182
5	1.5	0.724547	0.607794	0.004089	2.858881	1.953280	1.464
5	2	0.626090	0.587481	0.010584	3.020600	1.874374	1.612
3	1	0.706089	0.653302	0.063307	2.542182	2.298690	1.106
7	1	0.762339	0.642364	0.054180	2.604026	1.896347	1.373
10	1	0.785787	0.642364	0.054180	3.056900	1.896347	1.612

The coincidence overlap and the optical quantities derived from it are listed in [Table 4](#).

Table 4: Coincidence overlap, oscillator strength, and radiative lifetimes. Numerical-resolution estimates are omitted because they are small compared with the geometry-dependent variations.

a	c	$ M_X ^2$	f_X	τ_X (ps)	τ_{XX} (ps)
5	0.5	4.167	53.48	2.548	0.637
5	1	6.094	87.75	1.954	0.489
5	1.5	8.351	122.99	1.459	0.365
5	2	8.543	126.82	1.437	0.359
3	1	4.294	61.70	2.767	0.692
7	1	8.151	117.45	1.462	0.366
10	1	11.169	161.02	1.068	0.267

The final numerical estimator was selected through the sequence summarized in [Table 5](#). These trials are not additional uncertainty samples; they document why the complete correlation expression with only the global rotation removed was adopted for production.

Table 5: Summary of the numerical validation sequence at $(a, c) = (5, 1)$.

Trial	Main observation
Reduced budget convergence	The binding changed from -0.00893 to $+0.00240$ Ry between 10^6 and 2×10^6 points; the worst independently integrated biexciton identity spread was approximately 2.4%.
Dithered adaptive partitions	Three runs with nonzero bisection dithering retained spreads of approximately 1.9–2.7%; randomized partitioning did not remove the problem.
Complete four-angle estimator	All reflection and particle-exchange reductions were removed. The estimator retained a redundant global rotation and displayed larger budget drift.
Global-rotation-fixed estimator	One global angle was removed, while the other angles retained their full ranges. This formulation agreed with the complete four-angle estimator within the observed resolution at lower computational cost.
Staged ordinary minimization	A low-budget optimization followed by high-budget readout remained vulnerable to changes in candidate ordering.
Safeguarded pattern search	Candidate moves were required to improve the energy at both 10^6 and 2×10^6 points and to exceed the observed budget drift. This became the production protocol.

For reproducibility, the machine-readable final table is distributed with the manuscript source as the file

`data/main-state-final-summary-Ry.csv`

The underlying project repository additionally retains the Wolfram Language production drivers, checkpointed WXF result stores, per-candidate and per-integral CSV records, the plotting code, and the vector PDF figures. Each recorded integral includes its value, internal error estimate, elapsed time, warning text, and whether the point cap was reached.

The numerical history should be interpreted in the following hierarchy: the final pattern-search stores define the optimized parameters; the final summary CSV defines the plotted numerical values; and the earlier reduced, dithered, and four-angle files are validation records rather than alternative production estimates.